

Albara Mehene

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Professional Experience

Software Engineer II, Booz Allen Hamilton, Lexington, MA

August 2025 - Present

- Maintain and extend C++ code in the radar signal processing pipeline; debug real-time data handling issues and tune performance for defense workloads.
- Built an internal workflow assistant for analysts (Python/FastAPI backend, Angular frontend) using Claude and Llama models, handling lookups, report, and file retrieval through a conversational interface.

Software Engineer II, Canon U.S.A., Cambridge, MA

September 2023 - June 2025

- Built a C++ API enabling real-time communication between GUI interfaces and medical robotic hardware that follow IEC 62304 standards, improving response time during clinical operations.
- Developed a service health manager for monitoring temperature, voltage, and current diagnostics on RTOS, helping prevent hardware failures.
- Developed OpenGL-based visualization for bronchoscopy imaging system, rendering real-time endoscopic camera feeds during clinical lung procedures.
- Automated CI/CD pipelines in Jenkins, taking deployment frequency from biweekly to multiple times per week and removing manual release steps.
- Optimized and modularized legacy codebase, enabling independent module compilation and reducing build times by 60%.

Software Engineer II, BAE Systems, Burlington, MA

January 2022 - September 2023

- Built a React/PostgreSQL dashboard tracking live network traffic between radio platforms replacing manual log inspection during sim runs.
- Developed new features and tools in C++ for aircraft control & a messaging system in Python used in real-time simulation runs.
- Maintained and extended software running on radio platforms using Python to translate coordinates and communicate between devices.
- Refactored and decoupled core legacy components to utilize newer developer tools, resulting in over 30% reduction in build time and faster testing.
- Led & migrated CI/CD environment from BitBucket to Gitlab to gain access to newer tools and functionality such as Docker.

Software Engineer I, Raytheon Technologies, Woburn, MA

June 2019 – December 2021

- Developed and maintained tools in C++ that helped improve functionality and efficiency by implementing a UI with additional features and capabilities for radar runs.
- Refactored and refined legacy code for better maintainability and in support of a migrating effort to newer technologies for the team.

Education

University of Massachusetts Lowell, Lowell, MA

- B.S in Computer Science, Graduated Spring 2019 | Lowell, MA

Skills

- **Programming Languages & Libraries:** C++, Python, JavaScript, React.js, C, SQL, NoSQL
- **Technologies:** Git, Jenkins, IoT, GPIO, SPI, I2C, RS-232, UART, wxWidgets, QT Widgets, OpenGL, [Node.js](#), PostgreSQL, Grafana, AWS, Docker, Bitbucket, Sonnet, Haiku, Llama 3 8B
- **Operating Systems:** Windows, FreeRTOS, Linux (Ubuntu, Debian, RedHat)